

Week 2 cyber security

Task 1:-

1. LAN — Local Area Network:- A LAN is a network that connects computers and other devices within a small geographical area, such as a room, home, school, office, or computer laboratory.

The computers connected together in a school computer laboratory form a LAN.

- Homes
- Schools and colleges
- Offices
- Small businesses

2. MAN — Metropolitan Area Network:- A MAN is a network that connects multiple LANs across a larger area, usually covering a city or a large town.

A university may connect its different campuses located across a city using a MAN.

- Universities with multiple campuses
- Government organizations
- City-wide networks

3. WAN — Wide Area Network:- A WAN connects networks over a very large geographical area, such as different cities, countries, or continents.

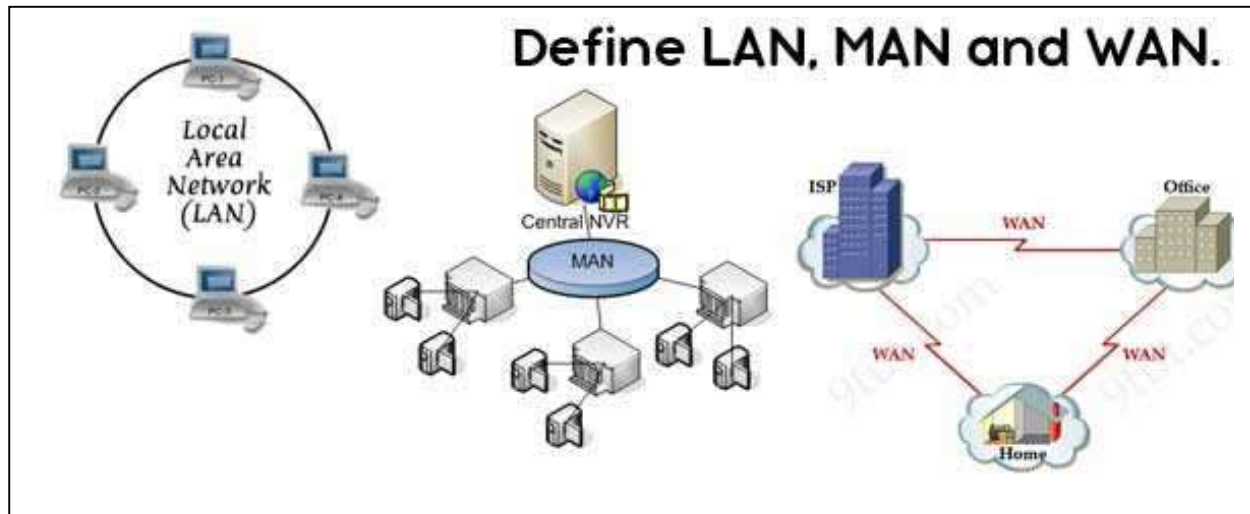
The Internet is the largest example of a WAN because it connects networks around the world.

- Internet
- International companies
- Banks
- Cloud services
- Organizations with offices in different countries

4. PAN — Personal Area Network:- A PAN is a very small network used to connect personal devices around an individual, usually within a few metres.

Connecting a smartphone to wireless earbuds using Bluetooth.

- Smartphones
- Smartwatches



Task 2:-

An **IP (Internet Protocol) address** is a numerical address assigned to a device on a network. It helps identify the device and allows data to be delivered to the correct destination.

Example: 192.168.1.10

An IP address can change depending on the network and how the device receives its address.

IPv4 stands for **Internet Protocol version 4**.

It uses a **32-bit address** and is normally written as four numbers separated by dots.

Example: 192.168.1.25

Each number can range from **0 to 255**.

A **private IP address** is used inside a local network, such as a home, school, or office network. It is not directly reachable from the public Internet.

Common private IPv4 ranges include:

10.0.0.0 – 10.255.255.255

172.16.0.0 – 172.31.255.255

192.168.0.0 – 192.168.255.255

Example: 192.168.1.5

A **public IP address** is an address used to identify a network or device on the public Internet. It is normally assigned by an Internet Service Provider (ISP).

MAC stands for **Media Access Control**.

A MAC address is a hardware/network-interface identifier associated with a network interface. It is normally represented as six groups of hexadecimal characters.

Example: A4:5E:60:12:AB:9A

It is primarily used for communication within a local network.

Difference Between IP Address and MAC Address

Feature	IP Address	MAC Address
Full form	Internet Protocol Address	Media Access Control Address
Purpose	Identifies a device/network location for IP communication	Identifies a network interface on the local network
Type	Logical/network-layer address	Link-layer/hardware identifier
Example	192.168.1.17	A4:5E:60:12:AB:9A
Can change?	Often can change	Usually associated with the network interface, though devices can use randomized/spoofed MAC addresses

Feature	IP Address	MAC Address
Used mainly for	Routing data between networks	Local network communication

Task 3:-

1. Router

A router connects different networks and forwards data packets between them. In a home network, it commonly connects the local network to the Internet.

Homes, offices, schools, businesses, and data networks.

example:

A home Wi-Fi router supplied by an Internet service provider.

2. Switch

A switch connects multiple wired devices within a local network and forwards Ethernet frames to the appropriate device.

Offices, schools, computer laboratories, and data centers.

example:

A 24-port Ethernet switch connecting computers and printers in an office.

3. Modem

A modem provides the connection between a local network and an ISP's access network by converting/signaling data according to the access technology being used.

Homes and offices that receive Internet service through technologies such as cable or DSL.

example:

A cable modem used to connect a home network to a cable Internet service.

4. Access Point:-An access point provides wireless network access to devices such as smartphones, laptops, and tablets.

Homes, offices, schools, hotels, airports, and public places.

example:

A ceiling-mounted Wi-Fi access point in an office.

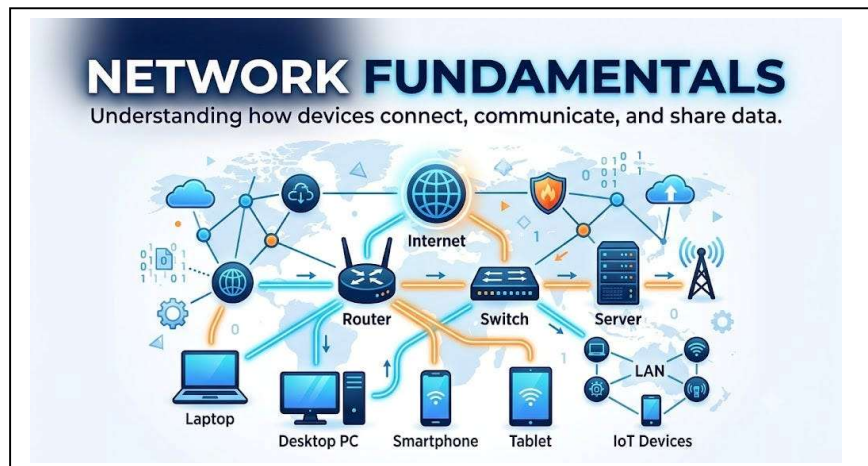
5. Firewall

A firewall monitors and controls network traffic according to security rules. It can help block unauthorized or unwanted connections.

Computers, home routers, offices, organizations, and data centers.

example:

A firewall built into a home router that blocks unwanted incoming connections.



Task 4:-

Protocol Full Form		Purpose	Real-World Example
HTTP	Hypertext Transfer Protocol	Transfers web resources between a browser and web server	Loading a webpage
HTTPS	Hypertext Transfer Protocol Secure	Provides HTTP communication protected using TLS encryption	Online banking website
DNS	Domain Name System	Converts domain names into IP addresses	Finding the IP address for example.com
TCP	Transmission Control Protocol	Provides reliable, ordered delivery of data	Downloading a file
UDP	User Datagram Protocol	Provides fast, connectionless transport without guaranteeing delivery	Live video or online gaming

HTTP

HTTP allows web browsers and web servers to communicate and exchange web resources.

Example:

A browser requests a webpage from a web server using HTTP.

HTTPS

HTTPS is HTTP protected with **TLS (Transport Layer Security)**. It helps protect data from being read or modified while it travels between the browser and server.

Example:

Entering information on an online shopping website.

DNS

DNS translates human-readable domain names into IP addresses that computers can use for network communication.

Example:

www.example.com



DNS lookup



IP address

TCP

TCP establishes a connection and provides reliable, ordered delivery of data. If appropriate, lost data can be retransmitted.

Example:

Downloading a document where accurate delivery is important.

UDP

UDP is a connectionless transport protocol with lower overhead than TCP. It does not guarantee delivery or ordering.

Example:

Real-time applications such as online gaming and some live audio/video traffic.

Task 5:-

When you type:

www.example.com

into a browser, several steps occur.

Step 1 — User

The user enters:

www.example.com

into the browser's address bar.

Step 2 — Browser

The browser determines that it needs to access the website and begins the process of locating the server.

Step 3 — DNS

The domain name needs to be translated into an IP address.

The computer sends a DNS query to a DNS resolver.

www.example.com



DNS Resolver



IP Address

Step 4 — Router

The computer sends network traffic toward its default gateway, usually the local router.

The router forwards the traffic toward the Internet.

Step 5 — Internet

The data travels through multiple networks and routers across the Internet toward the destination server.

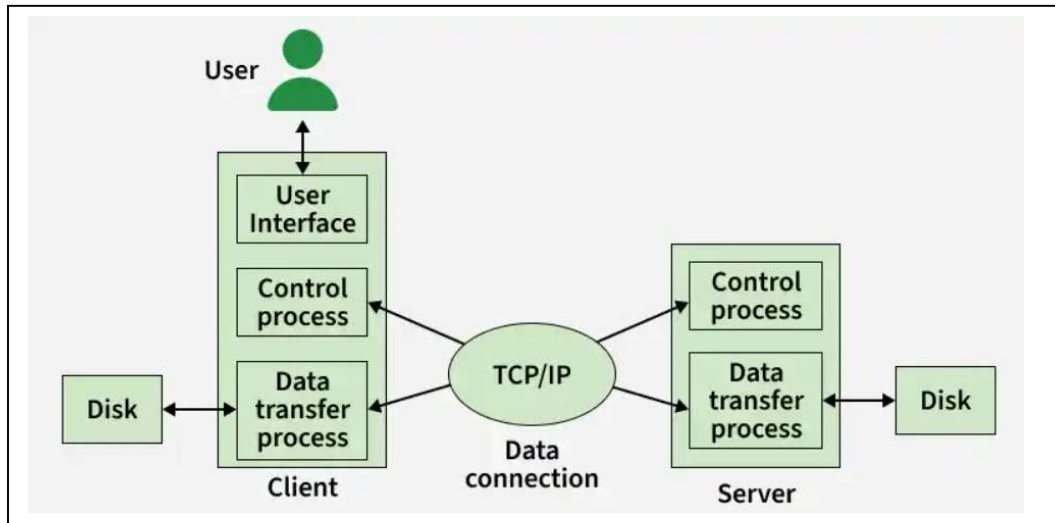
Step 6 — Server

The destination web server receives the request and processes it.

Step 7 — Response

The server sends the requested webpage/resources back through the network.

The browser receives the response and displays the webpage.



Task 6:-

10 Basic Practices for Securing a Home Wi-Fi Network

1. Use a **strong and unique Wi-Fi password**.
2. Change the **default administrator password** of the router.
3. Keep the router's **firmware updated**.
4. Use **WPA2 or WPA3** security instead of outdated security methods.
5. Avoid connecting to **unknown or suspicious Wi-Fi networks**.
6. Disable **WPS** if it is not needed.
7. Change the router's default network name if appropriate.
8. Use a **guest network** for visitors and untrusted devices.
9. Disable **remote administration** unless it is genuinely required.
10. Regularly check the router for **unknown connected devices**.

Explain Any 3 Practices in Detail

1. Use a Strong Wi-Fi Password

A weak password can be guessed or attacked more easily. A strong password should be long and difficult to guess. It should avoid easily available personal information such as names or birthdays.

Example:

Weak: password123

Stronger: BlueRiver!47Cloud#92

A unique password also prevents a password reused on another service from compromising the Wi-Fi network.

2. Keep Router Firmware Updated

Router manufacturers release firmware updates to fix security vulnerabilities, improve performance, and sometimes add new features.

Therefore, users should regularly check whether their router has a newer firmware version and install updates from the manufacturer's official source.

3. Use WPA2 or WPA3

WPA2 and WPA3 are wireless security standards designed to protect Wi-Fi communications. WPA3 provides newer security improvements where supported.

Users should avoid obsolete security methods such as **WEP**, which is no longer considered secure.